Yingrong Chen

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Education

University of Washington — Seattle, WA
M.S. in Physics (Part-time) — *Sep 2025 – Present*

Emory University — Atlanta, GA
B.S. in Chemistry (High Honors), Double Major in Computer Science (GPA: 3.987 / 4.0) — *Aug 2020 – Dec 2023*

Work Experience

Microsoft — Redmond, WA

Quantum Software Engineer 2 — *Jul 2025 – Present*

- Developed and evaluated **fault-tolerant quantum algorithms**, including Shor's algorithm, QPE, Hamiltonian simulation, block encoding, and quantum chemistry workflows, analyzing algorithmic performance and resource tradeoffs.
- Performed **quantum resource estimation and system-level analysis**, translating algorithms into logical-resource requirements and evaluating tradeoffs across designs and fault-tolerant architectures.
- Engineered **production-quality quantum software** with modular architectures, automated testing, CI/CD, documentation, reusable components, and notebooks for external and open-source users.
- Collaborated across **research, engineering, and product stakeholders** to translate ambiguous quantum research objectives into technical roadmaps, implementation milestones, demonstrations, and deliverables.

Contract Software Engineer — *Jan 2024 – Jun 2025*

- Designed a quantum algorithm for **sparse state preparation** (**patented**).
- Implemented quantum phase estimation, state preparation, and energy estimators in the open-source **qdk-chemistry** using Python, Qiskit, and Q#.
- Added **MPI** and **CUDA** support for ML-based exchange-correlation functionals in the open-source **skala** using C++.
- Implemented ddX solvation models and QM/MM workflows in C++ for in-house computational chemistry software.
- Designed scalable computational chemistry workflows on **Azure HPC**, integrating pharmacokinetic screening, consensus docking, and free-energy perturbation to process high-throughput data for 9M+ molecules targeting CDK12 protein.

Community & Research Experience

Quantum Open Source Foundation (QOSF) — Remote

Mentee — *Nov 2024 – Sep 2025*

- Implemented 10+ quantum arithmetic algorithms in Q#, including ripple-carry addition, Karatsuba multiplication, and windowed exponentiation. These serve as modular building blocks for quantum algorithms such as Shor's.
- Open-sourced the **QuantumArithmetic** library and integrated it into the Q# 1.15 release.

Fang Liu Lab at Emory University — Atlanta, GA

Research Student — *Jan 2022 – Dec 2023*

- Investigated how protein environments influence energy gap fluctuations in green fluorescent protein and its mutants using QM/MM simulations in Amber and computed their absorption spectra using TDDFT in TeraChem.

Jens Meiler Lab, Vanderbilt University & Rosetta Commons — Nashville, TN

Research Student — *May 2022 – Aug 2023*

- Studied phosphorylation effects on α-synuclein interactions with MHCII and TCR in Parkinson's disease using AlphaFold2, Rosetta and Amber.
- Integrated the BioChemical Library into Rosetta for protein-ligand energy evaluation and conformer generation in C++.

Selected Publications & Open Source Code

- Chen, Yingrong et al. *Clifford-efficient sparse state preparation for molecular wavefunctions.* arXiv:2608.20593 (2026).
- Baker, Nathan et al. *QDK/Chemistry: A Modular Toolkit for Quantum Chemistry Applications.* arXiv:2601.15253 (2026). [**microsoft/qdk-chemistry**]
- Luise, Giulia et al. *Accurate and Scalable Exchange–Correlation with Deep Learning.* arXiv:2506.14665 (2025). [**microsoft/skala**]
- Fedoriaka, Dmytro et al. *Quantum Arithmetic Algorithms: Implementation, Resource Estimation, and Comparison.* IEEE Quantum Week (QCE) 2025. [**fedimser/quant-arith-re**]

Awards & Honors

- **NSF Graduate Research Fellowship Honorable Mention** *2023*
- **Barry Goldwater Scholarship** *2022*
- **Emory Chemistry Academic Excellence** *2022*